

Department of Telecommunications and Artificial Intelligence
Faculty of Electrical Engineering and Informatics
Budapest University of Technology and Economics

Title: Agent-Orchestrated Statistical Modelling
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Abstract

Background: Building a statistical model for a tabular dataset is a long chain of decisions (data validation, exploratory analysis, splitting, preprocessing, encoding, model choice, tuning, evaluation and interpretation), and several of them can silently invalidate the result, for example data leakage or repeatedly evaluating candidates on the test set. AutoML systems automate the chain by brute-force search, while general-purpose LLM assistants give advice but neither execute the work reproducibly nor guarantee statistical validity.

Objectives: The goal is to design and implement a locally run web application in which an LLM agent designs and executes a comparison of several modelling approaches (models and methodologies such as encodings, feature strategies and preprocessing variants) on one user-supplied CSV file and an optional JSON data contract, then ranks them and selects a champion. Beside predictive performance, every run reports classical statistical inference: coefficients, standard errors, p-values, confidence intervals, VIF and residual diagnostics.

Methods: All computation lives in a small, typed, stateless tools exposed by a standalone Model Context Protocol (MCP) server built on pandas, scikit-learn and statsmodels. the LLM only sees artifact IDs and small JSON summaries, never raw data. A custom graph engine executes a declared 13-node pipeline: six shared nodes (intake, problem definition, exploratory data analysis, splitting, experiment design, inferential analysis), five nodes run once per candidate (preprocessing, feature engineering, training, evaluation, interpretation) and two final nodes (champion selection, artifact persistence). In the experiment-design node the agent proposes a reasoned shortlist of 3-5 candidates, each justified by an exploratory finding and including a mandatory interpretable baseline. Statistical honesty is enforced in code: one shared split for all candidates, a test set locked until champion selection, and the champion scored on it exactly once. The system supports automatic and guided (human-in-the-loop) modes, pause/resume, fork-based rewind and live event streaming to a React user interface. The LLM layer is provider-agnostic (LiteLLM with structured Pydantic outputs, DeepSeek by default).

Expected results: A working prototype of the application with automated tests for its main components, a generated standalone script intended to reproduce the selected model's results, a Markdown report with plots, and a first evaluation on a few public classification and regression datasets, comparing the agent-selected model with the baseline and discussing reproducibility, runtime, LLM cost and the limitations observed.

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Project Plan

Week number	Estimated Date	Planned Tasks
Week 1-3	07 – 25 September, 2026	Topic definition, initial discussion with supervisor, and work-plan preparation, architecture design, project foundation and MCP server foundation
Week 4	28 September – 02 October, 2026	Data intake, profiling (EDA) and data splitting tools
Week 5	05 – 09 October, 2026	Data cleaning, encoding, model training and evaluation tools
Week 6	12 – 16 October, 2026	First end-to-end pipeline without LLM, statistical inference tools
Week 7	19 – 23 October, 2026	Feature engineering, model interpretation, reporting and reproducibility tools
Week 8	26 – 30 October, 2026	MCP client and pipeline execution engine
Week 9	02 – 06 November, 2026	Parallel comparison of candidate models, LLM integration
Week 10	09 – 13 November, 2026	Agent decision logic for the pipeline steps, first agent-driven run
Week 11	16 – 20 November, 2026	Error recovery, run control, human-in-the-loop mode
Week 12	23 – 27 November, 2026	Backend REST API and live event streaming
Week 13	30 November – 04 December, 2026	Web user interface, evaluation on public datasets
Week 14 - 15	07 – 18 December, 2026	Finalization, written report submission, and oral presentation

Date: Budapest, 2026.10.01

Dukat Nandor

Student’s signature

Supervisor's signature